

A Compact 7 MeV SSMB-Compton Architecture for a Coherent 13.5 nm Lithography Source

Architecture II of the lumen EUV source family (rank 3); shared experimental gate
with the storage-ring SSMB undulator architecture

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Abstract

We specify a *metre-scale* EUV source architecture that combines steady-state microbunching (SSMB) with laser-Compton up-conversion. Inverse-Compton scattering reaches 13.5 nm from a 7 MeV electron beam ($\sim 455\times$ less energy than the undulator route), at a ~ 2.3 cm bending radius that yields a ~ 1 m tabletop ring; and because Compton up-conversion has no free-electron-laser gain condition, the source is *slice-energy-spread immune*. Adding SSMB makes it coherent and continuous-wave. The result is a compact, coherent, CW, slice-immune EUV source — the tabletop ambition reached by a different mechanism than the slice-spread-bound single-pass FEL. The binding residual is a Compton bandwidth-collection flux tax ($\sim 50\times$), recoverable only by stacking an IR enhancement cavity and energy-recovery current at their demonstrated maxima simultaneously. The architecture shares its single experimental gate — steady-state microbunching at 13.5 nm precision — with the storage-ring SSMB undulator architecture (companion paper I): one experiment de-risks both. The analysis is closed-form and reference-matched; it claims convergence on the same physics, not precedence.

1 Introduction

This is Architecture II of the lumen EUV source family (rank 3). It is the second of two *novel* architectures; the first (rank 2, paper I) is the storage-ring SSMB undulator. The two are genuinely different machines — here 7 MeV and inverse-Compton, there ~ 511 MeV and an undulator — so we document them separately, while making explicit that they reduce to one shared experimental gate (Section 4). This architecture is the more aggressive: it targets a true tabletop footprint.

2 Background: sidestepping the slice-spread wall

For an undulator FEL, gain collapses once $\sigma_\delta/\rho \gtrsim 1$ [1]; a 0.5 % laser-plasma beam never lases at 13.5 nm (H.031; demonstrated LPA-FEL floor ~ 25 nm [2]). Inverse-Compton scattering of a drive laser ($\lambda_x \approx \lambda_L/4\gamma^2$) has *no* Pierce gain condition: slice spread only broadens the line, so the architecture reaches 13.5 nm at very low beam energy (H.033). This is the mechanism the single-pass LPA-FEL (rank 6) cannot use and why that path is terminal-thin (H.038).

3 Architecture

At 7 MeV the bending radius is ~ 2.3 cm (1 T), giving a metre-scale ring (H.044). Combining SSMB microbunching (coherent CW) with Compton up-conversion at low energy yields

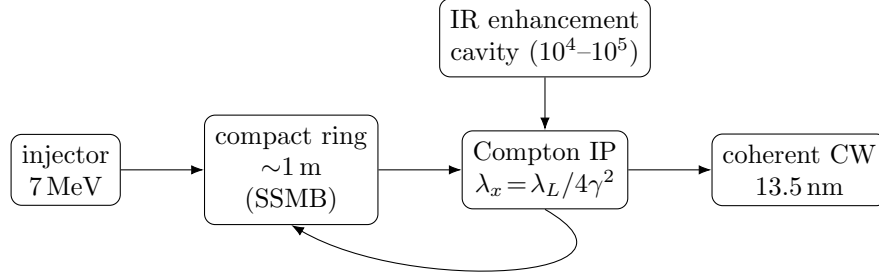


Figure 1: Compact SSMB-Compton architecture. A 7 MeV microbunched beam in a ~ 1 m ring scatters an IR drive laser (recirculated in an enhancement cavity) at the interaction point; Compton up-conversion to 13.5 nm is slice-spread immune.

a compact + coherent + CW + slice-immune source. The residual is a **flux tax**: the small Thomson cross-section puts the demonstrated Compton flux ($\sim 10^{13}$ ph/s) $\sim 7 \times 10^5$ below the $\sim 6.8 \times 10^{18}$ ph/s needed for 100 W (H.034). Recirculation closes most of it: an IR-wavelength enhancement cavity (finesse 10^4 – 10^5 — an EUV cavity is forbidden by multilayer $R \approx 0.70$, H.035) times energy-recovery average-current gain ($\sim 10^3$) gives $\sim 10^7$, exceeding the gap. But bandwidth-limited collection (a 2% in-band passband from the wide low- γ cone) derates flux $\sim 50 \times$ (H.036); it is recovered only by running finesse and current at their demonstrated maxima *simultaneously*. A brightness lever does not help — the λ – θ correlation is radiation kinematics, beam-independent (H.037, a recorded negative). A 2025 cavity-based compact EUV-litho proposal occupies this space [3].

Subsystems and secondary gates. Beyond the shared microbunching gate: the joint finesse $\times$ current maximum at EUV, the IR drive cavity integration, and Compton flux extraction.

4 The shared gate

The architecture rests on one unproven step: steady-state microbunching at 13.5 nm precision ($\sigma_z \approx 3$ nm), physics-allowed with coherent fraction $b^2 \approx 0.14$ ($b = \exp(-\frac{1}{2}(2\pi\sigma_z/\lambda)^2)$) (H.046). **The same gate binds the storage-ring SSMB undulator architecture (paper I), so one experiment de-risks both** (H.048). The route trade is exactly one axis: this architecture is metre-scale but pays the $\sim 50 \times$ Compton tax that the undulator route avoids at the cost of a larger ring. The cooling lever the lab derived closed-form (H.024) is what the latest SSMB frontier adds (OSC, [4]; H.050).

5 Related work and limitations

The SSMB mechanism was demonstrated by Deng et al. [5]; compact inverse-Compton sources are operational at hard X-ray and γ energies; a compact cavity-based EUV-litho source is proposed in He et al. [3]. All relations are first-order, closed-form, representative-parameter. The Compton flux tax is the binding gate after microbunching; 13.5 nm power from a compact SSMB-Compton ring is undemonstrated (concept stage). This is reference-match, not priority.

6 Reproducibility and future work

Closed-form evidence: lumen HYPOTHESES/cards/H_{033,034,035,036,037,044,046,048,050}.md + deterministic runners in state/h0*/; re-verify with python3 tool/qa.py. Decisive next

step: demonstrate steady-state microbunching at 13.5 nm ($b^2 \gtrsim 0.1$ over $\gtrsim 10^3$ turns), then the Compton flux-tax recovery at the joint operating point. If $\sigma_z \gtrsim 10$ nm or the derate exceeds the recoverable margin, the concept reverts to longer wavelengths (where $b^2 \rightarrow 1$).

References

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- [2] Hengyuan Xiao et al. Staged laser wakefield acceleration for saturated lasing of bandwidth-tunable free-electron lasers from euv to x-ray. *arXiv preprint arXiv:2603.28214*, 2026.
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